

Answer key

Name: _____

Roll No.: _____

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Amrita School of Computing, Bengaluru Campus
Department of Computer Science & Engineering
B.Tech CSE Third Semester (2025-26)
23ECE205 DIGITAL ELECTRONICS

QUIZ 02 (SET A)

Date: 23/09/2025

Duration: 15 minutes

Max. Marks: 10

CO	Course Outcomes
CO1	Realize a given expression in terms of basic building blocks.
CO2	Minimize a given logic expression.
CO3	Design combinational circuits
CO4	Design Sequential circuits.

Q. No.	01	02	03	04	05	06	07	08	09	10
Marks Obtained										
Total (out of 10)					Sign of student				Sign of faculty	

2
5
14
25+20
+3

Q1. Do the following conversions. Show steps of conversion (1 mark each)

[CO2, BTL2, 5 marks]

- a. $(01001100011100)_2 = (11434)_8$
b. $(4105765)_8 = (108BF5)_{16}$
c. $(163)_8 = (115)_{10}$
d. $(1080)_{10} = (10000111000)_2$
e. $(143)_5 = (48)_{10}$

ROUGH WORK

00010000100010111
 110101
 $8^2 8^1 8^0$
 $163 = 64 + 48 + 3$
 115

Q2. Add $(1110)_2$ and $(1011)_2$
BTL1, 01 mark]

[CO2,

1110
 1011
 11001

Q3. How many select lines are required for 32:1 MUX - 5

[CO2, BTL1, 01 mark]

Q4. Represent -32 in 2's complement notation using 8 bits.

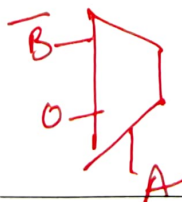
32
 00100000
 51110000

[CO2, BTL1, 01 mark]

Q5. Implement NOR Gate using 2:1 MUX.

[CO1, BTL2, 02 marks]

A	B	A+B
0	0	1
0	1	0
1	0	0
1	1	0



$2 \mid 1080$
 $2 \mid 545 - 0$
 $2 \mid 270 - 0$
 $2 \mid 135 - 00$
 $2 \mid 67 - 01$
 $2 \mid 33 - 01$
 $2 \mid 16 - 01$
 $2 \mid 8 - 0$
 $2 \mid 4 - 0$
 $2 \mid 2 - 0$
 $2 \mid 1 - 0$

Course Outcome (CO) / Bloom's Taxonomy Level (BTL) Mark Distribution Table

CO	Marks	BTL	Marks
CO1	00	BTL1	03
CO2	10	BTL2	05
CO3	00	BTL3	00
CO4	00	BTL4	02
-	-	BTL5	00
-	-	BTL6	00

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B.Tech CSE Third Semester (2025-26)
23ECE205 DIGITAL ELECTRONICS
QUIZ 02 (SET B)
Date: 23/09/2025

Max. Marks: 10

Duration: 15 minutes

CO	Course Outcomes
CO1	Realize a given expression in terms of basic building blocks.
CO2	Minimize a given logic expression.
CO3	Design combinational circuits
CO4	Design Sequential circuits.

Q. No.	01	02	03	04	05	06	07	08	09	10
Marks Obtained										
Total (out of 10)					Sign of student	Sign of faculty				

Q1. Do the following conversions. Show steps of conversion (1 mark each) [CO2, BTL2, 5 marks]

a. $(01111100011100)_2 = (17434)_8$

b. $(5104766)_8 = (1489F6)_{16}$

c. $(164)_8 = (116)_{10}$

d. $(1070)_{10} = (10000101110)_2$

e. $(134)_5 = (44)_{10}$

Q2. Add $(1010)_2$ and $(1110)_2$ [CO2, BTL1, 01 mark]

$$\begin{array}{r} 1010 \\ 1110 \\ \hline 11000 \end{array}$$

Q3. How many select lines are required for 64:1 MUX - 6 [CO2, BTL1, 01 mark]

Q4. Represent -23 in 2's complement notation using 8 bits.

$$\begin{array}{r} 23 \quad 168421 \\ 10111 \end{array} \quad \begin{array}{r} 00010111 \\ 11101001 \\ \hline 10000000 \end{array}$$

Q5. Implement NAND Gate using 2:1 MUX. [CO1, BTL2, 02 marks]

$$\begin{array}{c|c|c} A & B & A \cdot B \\ \hline 0 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{array}$$

If T-T is not there 1M.

ROUGH WORK

b) 101001000100111011010

1489F6

$$\begin{array}{r} 8 \times 164 = 1312 \\ 8 \times 8 = 64 \\ 8 \times 8 = 64 \\ 64 + 48 + 4 = 116 \end{array}$$

$$\begin{array}{r} 2 \mid 1070 \\ 2 \mid 535 - 0 \\ 2 \mid 267 - 1 \\ 2 \mid 133 - 1 \\ 2 \mid 66 - 1 \\ 2 \mid 33 - 0 \\ 2 \mid 16 - 1 \\ 2 \mid 8 - 0 \\ 2 \mid 4 - 0 \\ 2 \mid 2 - 0 \\ 2 \mid 1 - 0 \end{array}$$

$$\begin{array}{r} 5 \times 5 = 25 \\ 5 \times 5 = 25 \\ 5 \times 5 = 25 \\ 134 = 25 + 15 + 4 = 44 \end{array}$$

Course Outcome (CO) / Bloom's Taxonomy Level (BTL) Mark Distribution Table

CO	Marks	BTL	Marks
CO1	00	BTL1	03
CO2	10	BTL2	05
CO3	00	BTL3	00
CO4	00	BTL4	02
-	-	BTL5	00
-	-	BTL6	00